# Microbots

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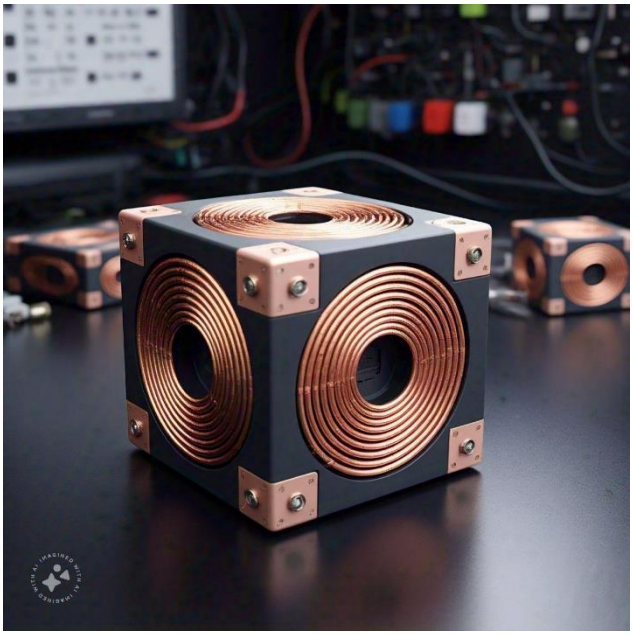
## Summary

It is a theoretical idea in which Millions of microbots designed to move as per our command and build 3D structures.

## Physical structure

The design is as follows, a cube with edges of order of few micrometres with coils on each face of the cube which will act as electromagnets, due to this individual microbots can attract and repel other bots. The coils can be printed on the particles using lithography. The microbots will have microchips printed on each face at the centre of the coils. The microchip will act as microcontrollers(brain) for each bot. Since these bots are small in size, they will have negligible mass but an ability to attract or repel other bots and make magnetic bonds.

Here is  an AI generated image of my idea:



# How will we power the microbots:

Powering the microbots with mini batteries or something can be challenging, however it's not needed. The microbots will not have any energy source of their own, since the microbots have many coils hence by using few of the them which will not be in use at a certain moment the bots can be powered, the microbots can be powered by electromagnetic induction (i.e an external oscillating magnetic field which will induce small current in the coils of microbots). The structure of bot can be made to resemble a capacitor which can store energy and use when needed. Hence powering the microbots will not be an issue as long as an external oscillating magnetic field is present.

# How these bots will be useful:

As said before the microbots will be a cube, we can use any material like metals, carbon,etc to increase the strength and variety. At each edge of each face a bare copper plate will be placed so that microbots can communicate with each others, transfer data or energy to the needed microbots. Due to this many microbots can come together and make a large bot with increased computing power improving usability.

The applications of this idea are infinite. These bots are like nanotechnology used by iron man in Marvel movies but a bit large in size. The applications are as follows.

1. Medical Applications

* Targeted Drug Delivery: Delivering medication directly to diseased cells or tissues minimizes side effects and increases treatment effectiveness.

* Minimally Invasive Surgery: Performing complex surgeries with minimal incisions, reducing patient trauma and recovery time.

* Disease Diagnosis: Detecting and diagnosing diseases at the cellular level, enabling early intervention and personalized treatment plans.

* Biopsy and Tissue Sampling: Collecting precise tissue samples for analysis, improving diagnostic accuracy.

2. Environmental Applications

* Pollution Remediation: Cleaning up contaminated environments, such as oil spills, by breaking down pollutants or absorbing them.

* Environmental Monitoring: Monitoring air and water quality, providing real-time data on pollution levels and other environmental factors.

3. Industrial Applications

* Precision Manufacturing: Assembling microscopic components, enabling the production of highly complex and miniaturized devices.

* Materials Science: Manipulating materials at the nanoscale, creating new materials with unique properties.

4. Other Potential Applications

* Defense and Security: Surveillance, reconnaissance, and targeted attacks.

* Space Exploration: Exploring and analyzing extraterrestrial environments, such as the surface of other planets or asteroids.

# How will we control these bots to operate and function as per our need:

The bots can be controlled by a technique I call duelplexing, due to this technique the bots have to use the coils for interactions only for few milliseconds in a second, so the bots can recharge their capacitors in the meantime and use stored energy at a particular instant. The technique is bit complex to explain. Hence another paper for that technique will be published later.

# Advantages and disadvantages:

Advantages

• these bots can act as building blocks and any structure can be made out of these bots.

• the strength of magnetic bond between two bots can be adjusted.

• since the energy is supplied to the bots externally the biggest problem is already solved.

• every individual bot can communicate with other bots and supply energy to others if needed.

• many bots can come together and form a processor of integrated computing power of each bot.

Disadvantages

• Manufacturing them with high precision can be challenging and costly.

# About me:

Hi, I am Nitin Vishwakarma and this is one of my many ideas, I am from India currently studying in 12th grade (high school). I wrote this to share and check the feasibility of this idea. I know this Idea is still incomplete and requires a lot of work and research but this idea can be used as a seed. Therefore I prefer to share this idea.

I would appreciate your feedback.

Kindly share your opinions on nitinfromindia4@gmail.com